```r
agg <- read_csv("../data/processed/store_type_a_train_aggregate_daily.csv")
```

4.3 ARIMA Method

Model Identification

```r
sales_ts <- ts(agg$Sales, frequency = 7)
auto_model <- auto.arima(sales_ts)
summary(auto_model)
```

```
## Series: sales_ts
## ARIMA(4,0,1)(2,1,0)[7]
##
## Coefficients:
##          ar1     ar2      ar3      ar4      ma1     sar1     sar2
##       1.0735  0.0047  -0.0166  -0.1645  -0.9669  -0.7877  -0.2346
## s.e.  0.0379  0.0489   0.0484   0.0350   0.0229   0.0376   0.0333
##
## sigma^2 = 1.077e+12:  log likelihood = -14278.73
## AIC=28573.47   AICc=28573.62   BIC=28612.19
##
## Training set error measures:
##                    ME    RMSE      MAE MPE MAPE      MASE        ACF1
## Training set 21175.98 1030280 646027.1 NaN  Inf 0.5758114 0.002753154
```

Automated model selection using auto_model identified ARIMA(4,0,1)(2,1,0)[7] as the initial candidate based on AIC minimization.

It indicates that there is:

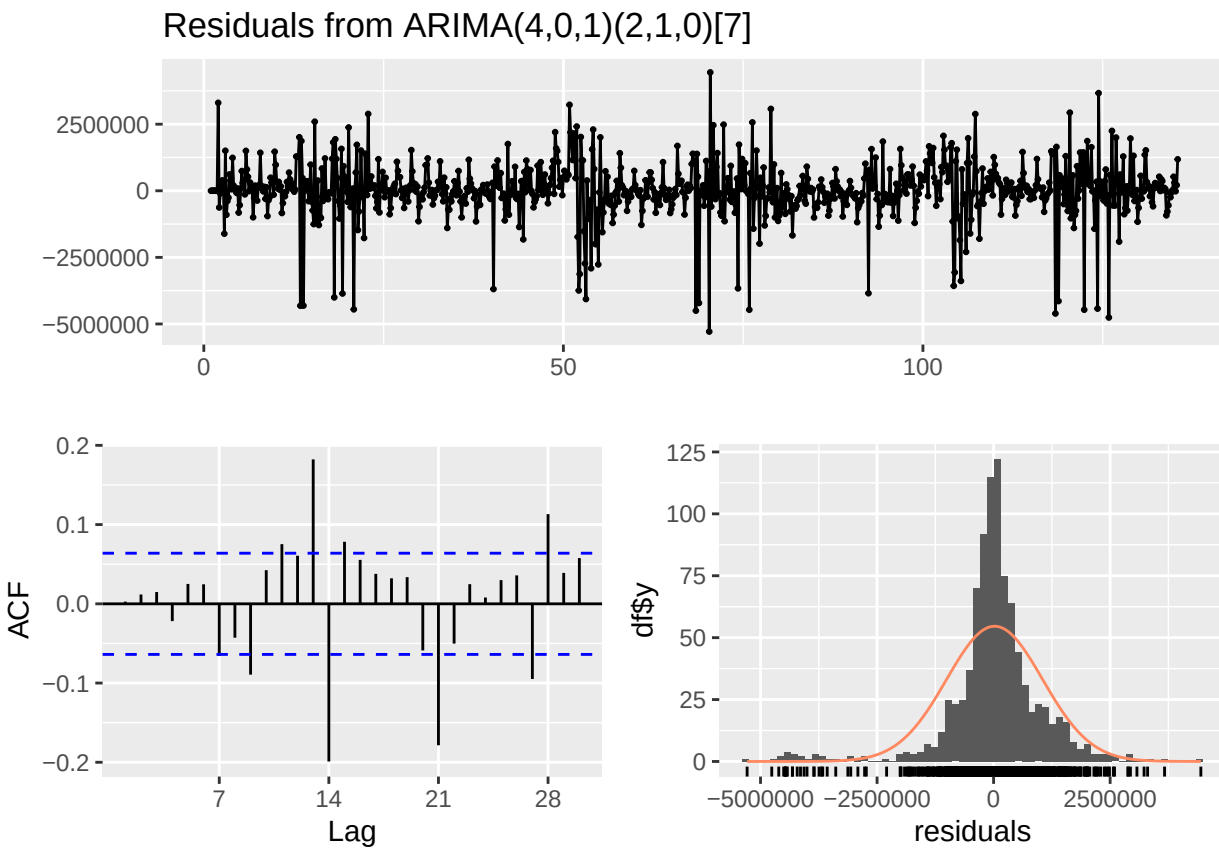
No non-seasonal differencing ($d = 0$), consistent with prior stationarity checks One seasonal difference ($D = 1$) to account for the strong weekly pattern in sales Both non-seasonal and seasonal autoregressive structure, since the model includes four AR terms and two seasonal AR terms A single non-seasonal moving average term to capture short-run shock correction

The coefficient estimates suggest that the first autoregressive term is the strongest among the non-seasonal AR components, while the second and third AR terms are quite small. The fourth AR term still contributes somewhat, and the moving average term is also large in magnitude. On the seasonal side, both seasonal AR terms appear important, especially the first seasonal term, which supports the presence of weekly dependence in the data.

Training performance metrics indicate a reasonable fit, with very low residual autocorrelation at lag 1 (ACF1 ≈ 0.003) and MASE < 1, suggesting the model performs better than a naive benchmark on the training set.

Residuals

```r
checkresiduals(auto_model)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(4,0,1)(2,1,0)[7]
## Q* = 95.835, df = 7, p-value < 2.2e-16
##
## Model df: 7.   Total lags used: 14
```

We check whether the fitted model has removed most of the structure from the series. If the residuals behave like random noise, then the model has captured the main signal well.

For ARIMA(4,0,1)(2,1,0)[7], the residuals are generally centered around zero, but there are still some noticeable large positive and negative spikes, suggesting the model does not fully explain all variation in sales. The residual ACF also shows several significant spikes, particularly around lags 13, 14, 21, 27, and 28, indicating that some weekly structure remains in the residuals. This suggests that although the model fits reasonably well, it does not completely capture the seasonal dynamics in the data.

Top plot: the residuals fluctuate around zero overall, which is good, but there are several large spikes and periods of higher volatility, indicating the presence of outliers and remaining unexplained variation.

ACF plot: multiple spikes fall outside the significance bounds, especially around lags 13–14 and 27–28, which are close to multiples of the weekly cycle. This indicates that some seasonal dependence is still present.

Histogram: the residuals are centered near zero and roughly bell-shaped, but with heavier tails due to extreme values, so the normality assumption is only approximately satisfied.

Overall, the residual diagnostics suggest that ARIMA(4,0,1)(2,1,0)[7] captures much of the main structure, but there is still remaining seasonal autocorrelation, so alternative models could be considered for further improvement.

Candidate Models

Candidate Models

The best AIC model is not always the best forecasting model, so we explore alternative specifications motivated by the residual diagnostics. In particular, the residual ACF showed remaining spikes at seasonal lags (around 13–14 and 27–28), suggesting that additional seasonal structure may not be fully captured. We also consider simplifying the non-seasonal AR terms since some higher-order coefficients are relatively small.

```
model1 <- auto_model
model2 <- Arima(sales_ts, order=c(4,0,1), seasonal=c(2,1,1))
model3 <- Arima(sales_ts, order=c(4,0,1), seasonal=c(1,1,0))
model4 <- Arima(sales_ts, order=c(2,0,1), seasonal=c(2,1,0))
model5 <- Arima(sales_ts, order=c(4,0,2), seasonal=c(2,1,0))
```

The reasons for considering these models are:

- **Model2**: Residual ACF showed remaining seasonal autocorrelation, so we increase the seasonal MA component ($Q = 1$) to better capture weekly effects
- **Model3**: Tests whether a simpler seasonal structure ($P = 1$) is sufficient instead of $P = 2$
- **Model4**: Since some higher-order AR terms are small, we reduce the non-seasonal order to $p = 2$ for a more parsimonious model
- **Model5**: Adds an additional MA term to capture short-term dependencies not fully explained in the baseline model

---

Models Comparison

```
AIC(model1, model2, model3, model4, model5)
```

```
##        df      AIC
## model1  8 28573.47
## model2  9 28350.66
## model3  7 28618.78
## model4  6 28621.32
## model5  9 28573.24
```

ARIMA(4,0,1)(2,1,1)[7]: 28573.47

ARIMA(4,0,1)(1,1,0)[7]: 28350.66 (lowest AIC)

ARIMA(2,0,2)(0,1,1)[7]: 28618.78

ARIMA(4,0,2)(1,1,1)[7]: 28621.32

ARIMA(4,0,2)(0,1,2)[7]: 28573.24

The baseline model, **ARIMA(4,0,1)(2,1,0)[7]**, provides a strong fit with low AIC. However, given the presence of residual autocorrelation at seasonal lags, alternative models may offer improvements in forecasting performance. Therefore, while AIC provides a useful starting point, model comparison using out-of-sample metrics such as RMSE is necessary to select the final model.